THE EOA PROGRAM, PART 0: THE ENGINEERING OF ALPHABETS AS AN OPEN FRONTIER AT THE INTERSECTION OF SYMBOLIC DYNAMICS, REPRESENTATION THEORY, AND DETERMINISTIC MACHINE LEARNING

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Abstract

We introduce the Engineering of Alphabets (EOA), a research program built on an observation that, as far as we can tell, current theories of representation do not predict. At the center of the program is a recursive operator we call beta. It takes the spoken names of alphabet letters and maps them into numerical sequences. When we run beta on the 26 letters of English under a bijective phonetic mapping, the sequences converge to a limiting constant ratio (LCR). For the English primary mapping, that ratio is about 3.306. Under other encodings we tested, it shifts: 3.037, 3.173, or 2.689.

Here is the strange part. The 26 sequences do not stay independent. They collapse into what looks like a 16-element basis. Build word vectors from that basis, and they land on a very low-dimensional subspace. We hypothesize, on the basis of our unreleased runs, that a recursive, tree-like generative structure drives the collapse; formalization and verification are pending (Section 11). In our own runs of the operator, these phenomena are stable and vary with the encoding. Independent testing is not possible yet, because the operator is not public. We plan to open it to collaborators under the academic collaboration terms described in Section 12.

This paper is Part 0 of the EOA Program. It offers no proofs and no conclusive experiments. Its job is to pose the questions that Parts I through V will try to answer, and to place those questions where they belong: at the intersection of symbolic dynamics [4], phonetic encoding theory, low-rank representation learning, and deterministic machine learning. We think EOA points to a real gap in the literature we have read: a class of deterministic string-to-sequence operators whose behavior is easy to observe experimentally but that, as far as we can tell, existing theories do not explain. The questions are answerable, but answering them will take tools that no single discipline currently has on its own.

KEYWORDS *Alphabet engineering, deterministic representation, symbolic dynamics, phonetic encoding, low-rank representation.*

1. WHAT IS THE ENGINEERING OF ALPHABETS?

The Engineering of Alphabets (EOA) is not a theory. It is a research program built on observations that, as far as we can tell, current frameworks do not predict. We first put the underlying observations on the public record back in 2010 [3], and this paper, along with the parts that follow, reports on where that work has led since.

The observations are as follows. Take the spoken names of the 26 English letters and feed them into a fixed deterministic recursive operator, beta. Run it 43 times. The result is 26 numerical sequences, and they behave in ways we still cannot explain:

- Every sequence converges to the same limiting ratio, but the actual value depends on the phonetic encoding. Under one bijective mapping, lambda is roughly 3.306. Switch to another mapping and it drops to 3.037, or 3.173, or 2.689.
- The 26 sequences are not independent. They collapse into what looks like a 16-element basis.
- Word vectors built from that basis end up on a subspace with low effective dimensionality.
- Change the encoding of a single letter while leaving the rest untouched, and the global ratio appears to break.

We want to be clear: these are observations, not theorems. They still need independent verification and theoretical grounding. We do not know why the operator converges. We do not know why the ratio is shared across all letters when the encoding stays consistent. We do not know whether the 16-element basis is a phonetic universal, a graph-theoretic constraint, or just a numerical coincidence. We do not know if the dimensionality collapse is a bug we should fix or a feature we should exploit. And we honestly do not know whether any of this matters for natural language processing, information theory, cryptography, or the design of future symbolic systems.

EOA is simply the program that chases these questions down. The problem draws on several disciplines, each of which contributes tools to a different part of the analysis:

- Symbolic dynamics [4]: provides the formal framework for studying substitution systems (Section 2).
- Phonetics and encoding theory: characterize the mappings that define the alphabet as a structured input (Sections 3 and 4).
- Combinatorics, representation theory, and linear algebra: provide the concepts needed to analyze the basis structure and the dimensionality of the resulting representations (Sections 4 and 5).
- Cross-linguistic comparison: informs the investigation of possible universals (Section 7).
- Compression and deterministic alternatives to learned embeddings: motivate the application-related questions discussed in Section 8.

To the best of our knowledge, this combination of problems has not been treated systematically within a single disciplinary framework. That is exactly why the program is necessary. Within this paper, the term 'deterministic machine learning' denotes the construction of representations by fixed rules, with no learning step: the same input always yields the same output, and no parameters are fitted to data. EOA is such a construction.

1.1 Why Part 0?

Scientific papers usually present answers. This one presents questions. The reason is strategic: the EOA Program spans multiple disciplines, and each discipline will approach the problem with different tools. By stating the questions before the answers, we allow researchers to engage with the problem on their own terms, without being constrained by the vocabulary or assumptions of our forthcoming empirical papers.

Part I [1] will present the sequences, the basis, and the dimensionality collapse. Part II [2] will present the encoding-dependence of the limiting ratio. Parts III, IV, and V will address transients, downstream behavior, and the operator itself. But none of those papers will answer the program's central question: what kind of system is this?

That question belongs to the community. This paper is the invitation.

1.2 The Structure of This Paper

The questions below are organized by the domain of knowledge they challenge. Each section poses problems that, as far as we can tell, the literature we have read does not answer. A few of these questions might have quick answers. Others look like they will need mathematical frameworks that nobody has built yet. Some could even collapse under scrutiny. They might be bad assumptions dressed up as questions. But every one of them is real.

The sections are:

- **Section 2** looks at the operator itself, from the perspective of mathematics and symbolic dynamics.
- **Section 3** tackles the limiting constant ratio, drawing on encoding theory, phonetics, and inverse design.
- **Section 4** explores the basis structure through combinatorics and group theory.
- **Section 5** asks if dimensionality collapse is a bug or a feature, via representation theory and linear algebra.
- **Section 6** examines the transient regime from an information-theoretic angle.
- **Section 7** steps outside English to consider cross-linguistic universals in linguistics and phonology.
- **Section 8** surveys the application landscape: computer science, cryptography, and current AI trends.
- **Section 9** takes up reverse engineering and target design (applied mathematics).
- **Section 10** asks how staged disclosure fits into philosophy of science.
- **Section 11** follows structural hints and unseen dynamics into fractal geometry and formal language theory.
- **Section 12** is a call for collaboration.
- Sections 13 and 14 close out with the program roadmap and concluding remarks.

1.3 Summary of Open Questions

To help you find the questions in your own field, the table below summarizes all 43 questions in one line each.

Table 1. Summary of Open Questions

Q#	Domain	What it asks
2.1	Operator	What operator class produces this convergence: substitution system, linear recurrence, contraction mapping, or none of these?
2.2	Operator	Why does a least-squares linear fit of the recurrence matrix fail? Where does the nonlinearity actually live?
2.3	Operator	Would random symbols or number words still converge to a shared ratio, or is alphabet-specific encoding required?
2.4	Operator	Is the transient regime monotonic, oscillatory, or chaotic, and does it survive at infinite precision?
3.1	Limiting ratio	Is λ a computable function of the encoding, predictable without running the operator?
3.2	Limiting ratio	Which encoding property actually determines λ: length, frequency, phonetic features, or coupling structure?
3.3	Limiting ratio	Is λ a global invariant of the alphabet, or does it depend on presentation order?
3.4	Limiting ratio	Is there a continuous mapping from phonetic space to λ-space, allowing λ to be

Q#	Domain	What it asks
		targeted by design?
3.5	Limiting ratio	Is there a universal ceiling on λ, or can an encoding push it arbitrarily high?
4.1	Basis	Why does the 26-letter alphabet collapse to exactly 16 basis sequences?
4.2	Basis	What property of the letter-name strings, not the phonemes, determines group membership?
4.3	Basis	Does the basis restructure under an alternate encoding, or stay fixed at 16 with different groupings?
4.4	Basis	Is the basis a unique fingerprint of the encoding, or can two different encodings share one?
5.1	Dim. collapse	What is the precise, metric-backed definition of the apparent rank-1 collapse?
5.2	Dim. collapse	Is the collapse a limitation to overcome, or a design feature of the representation?
5.3	Dim. collapse	Does the 43-term vector carry any information beyond a single scalar?
5.4	Dim. collapse	Can a guaranteed-collapsed representation be exploited for adversarial robustness?
6.1	Transient	How much letter-specific information survives in the detrended residual?
6.2	Transient	Are the transients structured or noisy, and do they converge to a secondary pattern?
6.3	Transient	Can the transient structure be recovered from the 16-element basis alone?
6.4	Transient	Would a truncated, non-collapsed vector (terms 1–20) be more useful than the full 43-term one?
7.1	Cross-linguistic	Do non-alphabetic scripts (syllabaries, abugidas, logographies) show analogous convergence?
7.2	Cross-linguistic	Is the tight Semitic cluster a structural property of abjads, or just a small-sample artifact?
7.3	Cross-linguistic	Would a synthetic alphabet with no phonetic history still produce a basis and an LCR?
7.4	Cross-linguistic	Is there a relationship between alphabet size and λ, and how would we test it properly?
8.1	Applications	Could a fully characterized operator serve as a useful cryptographic primitive?
8.2	Applications	Is there any downstream task where a low-rank, semantically empty representation actually helps?
8.3	Applications	Does determinism matter more than dimensionality in any real operational scenario?
8.4	Applications	Could truncated EOA vectors outperform the full vector on spelling, anagram, or OCR tasks?
8.5	Applications	Who would pay for a representation that is semantically empty but algebraically rigid?
8.6	Applications	Can EOA-43 act as a zero-training structural prior or fixed initialization for learned embeddings?
8.7	Applications	Can the dimensionality collapse be harnessed as a principled model-compression scheme?
9.1	Reverse design	Can we search for encodings whose λ approximates *π, e,* or the golden ratio?
9.2	Reverse design	What is the reachable set of λ values, and does it intersect mathematically significant numbers?
9.3	Reverse design	Can an encoding be designed so its sequence terms encode primes at specific positions?
9.4	Reverse design	What constraints govern inverse design? Must the encoding stay phonetic and bijective?
9.5	Reverse design	What would a targetable λ actually enable: periodicity, self-similarity, cryptographic use?
10.1	Disclosure	Can claims about a partially characterized operator be independently verified pre-disclosure?
10.2	Disclosure	Would full disclosure resolve these questions quickly, or are they hard regardless?
10.3	Disclosure	What precedent exists for staged disclosure in mathematics (RSA-129, AlphaFold)?
10.4	Disclosure	How should priority be adjudicated when the phenomenon is public but the mechanism is not?
11.1	Structural	Does the generation process exhibit a self-similar branching structure, and can its geometry be formally characterized?
11.2	Structural	Can the tree-like propagation be characterized by a finite grammar (e.g., L-system), and would such a grammar explain the basis and the collapse?

2. THE OPERATOR: WHAT GENERATES CONVERGENCE?

Beta takes a string, for example a letter name, and turns it into a sequence of real numbers. The full definition of the operator is not yet public. Section 10 explains why. In this paper we report the behavior we have observed in our own implementation. The ratio between consecutive terms settles in its decimal places. The further we move along the sequence, the more decimal places agree. In our 64-bit runs, the ratios agree to about 14 decimal places after the 43rd term, with some sequences settling earlier.

This is worth emphasizing: there is no gradient descent here. No loss function, no training data, no randomness. The convergence is baked into the recurrence itself.

Question 2.1. What kind of deterministic string-to-sequence operator behaves like this? Is beta a substitution system [4, 5], a coupled linear recurrence, a contraction mapping, or something else entirely? Maybe it is something that does not fit any of those boxes?

Question 2.2. If beta were a simple linear recurrence, we should be able to fit a matrix M by least squares. But the fit comes back inconclusive [2]. So where does the nonlinearity actually live? In the encoding step? In the recurrence? In the coupling between letters? Or somewhere we have not looked yet?

Question 2.3. Would the operator still converge to a shared ratio if it were applied to random symbols or to number words instead of letter names? If it does, the phenomenon may be formal rather than phonetic. If it does not, the alphabet-specific encoding is structurally necessary. This experiment is directly testable.

Question 2.4. Is the convergence monotonic, oscillatory, or chaotic in the transient regime? If we had infinite precision, would the transient structure disappear, or does it encode genuine letter-specific information that survives even in the limit?

3. THE LIMITING CONSTANT RATIO: WHY 3.306?

For one English bijective mapping, the LCR is approximately 3.306. For other bijective mappings, it is 3.037, 3.173, and 2.689. Throughout this paper these values are truncated to three decimal places for readability; our 64-bit runs resolve the ratio to about 14 decimal places (Section 2). These values are not universal constants. They appear to be functions of the encoding. The modified mappings, and the hypotheses H1 through H7 that motivate them, are defined in Part II [2].

Determining the full-precision value of the LCR, and whether it has an exact characterization such as an irrational number, is a distinct study that we do not attempt here. It requires higher-precision computation, for example 128-bit or arbitrary-precision arithmetic, and a careful treatment of the operator's disclosure constraints. We intend to pursue this study with interested collaborators; Section 12 describes the terms.

Question 3.1. Is lambda a computable function of the encoding? That is, given a phonetic mapping (a set of letter-name strings), can one predict lambda without running beta?

Question 3.2. If the answer to 3.1 is yes, what properties of the encoding determine lambda? Is it the mean letter-name length, the character-frequency distribution, the phonetic-feature profile, the graph-theoretic structure of cross-letter dependencies, or a joint interaction of all of these?

Question 3.3. If the answer to 3.1 is no, does lambda depend on the order in which the encoding is presented to the operator? In other words, is lambda a global invariant of the entire alphabet, or does it emerge from a specific sequential coupling?

Question 3.4. The English modified #1 encoding shifts lambda from 3.306 to 3.037, a substantial change. Arabic yields 2.751 and Hebrew 2.710, while Greek yields 4.128. Is there a continuous mapping from phonetic-space to lambda-space? Can we design an encoding whose LCR approximates a target value to arbitrary precision?

Question 3.5. If lambda is encoding-dependent, is there a universal bound? Can an encoding produce an arbitrarily large lambda, or does the operator impose a finite ceiling?

4. THE BASIS: WHY 16?

Preliminary observations suggest that the 26 letters of the English alphabet collapse to 16 unique sequences. Three groups appear to share sequences: (B, D, P, T, V, Z), (F, L, M, N, S), and (I, R). The remaining 13 letters appear to be singletons.

Question 4.1. Why 16? Is this number a function of the English alphabet size (26), or would a different alphabet size yield a different basis cardinality? If we added a 27th letter to English, would the basis grow to 17, or would the new letter slot into an existing group?

Question 4.2. The groupings do not correspond to any standard phonetic classification. B, D, P, T, V, Z are not all stops, not all fricatives, and not all voiced. F, L, M, N, S share no obvious articulatory feature. I and R are acoustically distant. What property of the letter-name strings (not the phonemes) determines membership in a group?

Question 4.3. Is the apparent 16-element basis a property of the English primary encoding only? If we apply the English alternative encoding, does the basis restructure into a different set of groupings, or does it remain 16 with the same letters grouped?

Question 4.4. If the basis is a fingerprint of the encoding, then two different encodings either produce different bases or the same one. If a second encoding produces the same basis, the basis is not a unique identifier. If every encoding gives a different basis, the basis is a unique signature, and it should be possible to invert it.

5. DIMENSIONALITY COLLAPSE: A BUG OR A DESIGN?

Because all basis sequences appear to converge to the same geometric rate, word vectors built from them seem to collapse to a very low-dimensional subspace, possibly effective rank 1.

Question 5.1. Preliminary observations suggest the vectors may collapse to a very low-dimensional subspace, but what is the precise definition of this collapse? Is it exact rank 1, numerical rank 1 under a specified tolerance, or an artifact of the construction formula? What metric (singular value ratio, cumulative variance, Frobenius approximation error) should be used to define “effective rank” in this context?

Question 5.2. If the collapse is a fixed property of the operator, is it a failure of representation or a feature of compression? In other words, does this mean letter-name strings contain only 1 bit of independent geometric information, or are we simply using the wrong decoder?

Question 5.3. If EOA-43 vectors collapse to a single direction, they are essentially just a weighted sum of letter positions and alphabetical indices. In that case, why does the operator produce 43 numbers instead of just one? Is there any information in those 43 numbers that one number cannot capture? This reduction is a hypothesis; we have not yet verified it.

Question 5.4. Can dimensionality collapse be exploited? If a representation is guaranteed to live on a line, can it be used to construct adversarially robust classifiers (where perturbations orthogonal to the line are harmless), or does near-chance accuracy on semantic tasks prove that the line points in a meaningless direction?

6. THE TRANSIENT: WHERE IS THE REAL INFORMATION?

Terms 1 through 38 appear to be letter-specific. They deviate from the geometric trend. They may be the only place where the 16 basis vectors differ.

Question 6.1. Here is something we can actually test. Detrend the sequences by dividing each term by lambda^n, and what is left is the transient. How much information is in that residual? Enough entropy to tell all 26 letters apart, or does it only distinguish the 16 basis groups? That distinction matters.

Question 6.2. Are these transients actually structured, or just noise? Do they settle into some secondary pattern, or do they stay chaotic through all 43 terms? And if there is a secondary ratio hiding in there, does it shift when we change the encoding, just like the primary ratio does?

Question 6.3. We keep wondering if the real value of this operator is buried in the transients, not the convergence. Can we recover that transient structure from the 16 basis vectors alone? Or do we need the full recurrence? This boils down to a simpler question: is the basis just a lossy compression of the transient, or is it something entirely separate?

Question 6.4. Suppose the transients really do encode letter identity. Could we build a representation that does not collapse by using only terms 1 through 20, throwing away the converged tail entirely? We have no idea if that would work, but it is worth trying.

7. CROSS-LINGUISTIC UNIVERSALS: BEYOND ENGLISH

We have run preliminary tests across seven encodings and four languages. The LCR shifts each time. The basis structure probably shifts too, but we honestly do not know how far this goes.

Question 7.1. What happens when we leave the alphabet entirely? Do syllabaries, abugidas, and logographies show anything analogous? Can beta even digest the names of hiragana, katakana, or Devanagari characters? If it can, do they converge to a shared ratio like the alphabets do? Or does the whole thing fall apart once we step outside the alphabetic paradigm?

Question 7.2. Here is a pattern that might mean something, or might mean nothing. Arabic sits at 2.751 and Hebrew at 2.710. English primary is at 3.306 and Greek jumps to 4.128, giving a much wider Indo-European spread. Is that tight Semitic grouping real? Maybe it comes from something structural, like consonantal roots, vowel omission, or how abjads work. Or maybe we just do not have enough data points to tell signal from noise.

Question 7.3. There is a direct way to test this. Build a synthetic alphabet from scratch, with no real phonetics and no natural language history, and run beta on it. Do a basis and an LCR still appear? If they do, the phenomenon is purely formal: it does not care where the strings come from. If they do not, then the phonetic grounding really matters. One condition matters for the test to be fair: the synthetic strings should match the natural ones in length and in the characters they use.

Question 7.4. Does alphabet size predict lambda? Using the primary bijective mapping for each language, English has 26 letters and lambda $\approx$ 3.306. Hebrew has 22 and lambda $\approx$ 2.710. Greek has 24 and lambda $\approx$ 4.128. Arabic has 28 and lambda $\approx$ 2.751. Four data points, one per language, is not a regression. It is a guess. What would a proper experimental design look like? How many alphabets, natural or synthetic, would we need to test before we could even frame the hypothesis seriously?

8. THE APPLICATION LANDSCAPE: CIPHER, COMPRESSION, AND AI

EOA-43 is not a word embedding in any conventional sense. But its deterministic, algebraic rigidity may have value in domains where semantics is not the goal. Does it carry any semantic content? How does it perform on standard benchmarks? These are open questions that Parts IV and V will address.

Question 8.1. If we fully understand how the operator works, could its string-to-sequence mapping be useful for keeping information secure? The apparently linear behavior of the recurrence suggests the mapping is probably easy to invert once the operator is public, which would be a major limitation for cryptography. Whether any useful property survives that limitation is a question we have not analyzed.

Question 8.2. Is there any real use for a representation that carries no meaning but is algebraically rigid? Or is it only a mathematical curiosity?

Question 8.3. EOA-43 uses 2.69 KB (the 16 basis sequences of 43 numbers each, stored as 32-bit floats), the same as a random projection. Does its determinism ever matter more than its size? For example, in reproducible research, or in deployment on any device without retraining?

Question 8.4. If the transient terms (Section 6) contain recoverable letter-specific information, could a truncated EOA representation (terms 1 through 20 only) outperform the full 43-term vector on character-level tasks such as spelling correction, anagram detection, or OCR post-processing?

Question 8.5. A representation that carries no semantic content but is algebraically rigid needs a concrete use before it has value. Is the value in the vectors themselves, or in the operator that generates them? We are not sure there is a practical market for this representation, and we have not studied it.

Question 8.6. The current AI conversation keeps circling back to compact representations, on-device inference, and deterministic reproducibility. EOA-43 fits this pattern. We call it a zero-compression representation: the vector is small from the start, produced by a fixed mapping from the letters' own information, with no training step, no stored model, and nothing to decompress at runtime. It is 2.69 KB and needs no training. Could it serve as a structural prior in such a pipeline? Could it seed learned embeddings as a fixed initialization, shrinking the parameter count downstream? Maybe, but we have not tested any of this yet.

Question 8.7. What if the collapse could be treated as a feature? One untested idea is to take a 768-dimensional BERT embedding and project it onto the EOA line. If the loss on a specific task stayed minimal, that would suggest a principled compression scheme. If not, we would simply be throwing information away. We have not tested this.

9. REVERSE ENGINEERING AND TARGET DESIGN

We know lambda depends on the encoding. So can we flip that dependency? Can we design an encoding to hit a target lambda, rather than discovering what lambda an encoding happens to produce?

Question 9.1. The math is clear: for a fixed finite alphabet, the set of possible encodings is countable. That means the set of reachable LCR values is at most countable too. But "countable" and "practically searchable" are different things. Could we actually hunt for an encoding whose LCR lands arbitrarily close to pi? To e? To the golden ratio? We do not know if the search space is sparse or dense.

Question 9.2. Even if we cannot hit pi exactly, we should at least be able to characterize what is reachable. What does the image of the encoding-to-LCR mapping actually look like? A related question is whether that image intersects any numbers mathematicians already care about.

Question 9.3. Can we push further and tune the recurrence itself? Specifically, could we design an encoding so that the sequence terms hit prime numbers at specific positions? That would mean understanding whether the operator can be coaxed into generating number-theoretic sequences. That is something we have not even begun to figure out.

Question 9.4. Say inverse design is possible. What are the rules? Does the encoding have to stay bijective and phonetic, or would any consistent string assignment work? And how would we search? Gradient-free optimization might struggle because string space is discrete. Does this become a combinatorial nightmare, or is there a smarter way in?

Question 9.5. What would a targetable LCR actually buy us? We can imagine a few possibilities, but they are only guesses. A lambda close to pi might give vectors with a circular or repeating shape. A lambda near the golden ratio might give self-similar patterns. A lambda near a prime might be useful for keys or error-correcting codes. We list these simply to show why controlling the target would matter. None of this may work out.

10. THE EPISTEMOLOGY OF STAGED DISCLOSURE

Beta is still under active investigation. We have not published the full recurrence yet, and that decision raises uncomfortable questions about how science ought to move forward when the mechanism behind the data is still private.

Question 10.1. Can anyone actually verify a claim about an operator they cannot fully see? We have documented the sequences and their properties in detail, but we are genuinely unsure whether that is sufficient for mathematical verification. Does a real proof require access to the recurrence itself, or can the behavior stand alone?

Question 10.2. Suppose we published the full characterization tomorrow. Would these questions start falling into place within a year, or would they remain stubbornly open because the mathematics is genuinely hard? we keep coming back to this: is our current ignorance just a side effect of holding things back, or would the problem resist us no matter how much we shared?

Question 10.3. We are hardly the first to release results before methods. RSA sat on RSA-129 for decades as a challenge. DeepMind put AlphaFold structures into the world before the code followed. Does that history excuse what we are doing, or is EOA a completely different animal? we do not have a clean answer.

Question 10.4. The priority question is this. If someone independently reconstructs the recurrence without ever seeing our operator, who owns that discovery? When the empirical phenomenon is public but the generative mechanism is not, how do we fairly assign credit? We do not know the protocol.

11. STRUCTURAL HINTS AND UNSEEN DYNAMICS

Beyond the numbers we have reported, we have one further observation, which comes from our own runs of the not-yet-public operator and still needs independent confirmation. Start with a single symbol, and the expansion branches out along multiple paths. Eventually it settles into a finite, closed set, but that set does not stop producing descendants. They keep emerging, infinitely. That behavior is hard to reconcile: a closed set that is also inexhaustible. It points to some recursive, tree-like structure sitting behind the basis collapse, though we have not been able to formalize it yet. We raise it here because it might be the thread that connects the numerical results to an actual generative mechanism.

Question 11.1. Does the generation process exhibit a recursive, self-similar branching structure? Does this indicate a formal fractal geometry, an L-system grammar, or a graph-theoretic attractor? If so, what is its branching ratio, and how does it relate to the limiting constant ratio λ?

Question 11.2. If the branching turns out to follow an L-system, can we write down the exact rules that generate it? And would those rules explain why the basis has exactly 16 elements and why the dimensionality collapses? We want to know if they can do more than confirm that it happens. We want them to show the mechanism behind it.

12. CALL FOR COLLABORATION

The questions posed in this paper are not solvable by a single researcher or a single discipline. They require expertise that spans pure mathematics, computational linguistics, representation learning, cryptography, and phonetic theory. We therefore issue an open call for collaboration.

We are prepared to offer the following to qualified collaborators:

- Access to preliminary sequence data under standard academic agreements.
- Access to computational tools via a secure notebook environment that allows testing of arbitrary encodings without exposing the underlying recurrence.
- Co-authorship on papers that produce correct, reproducible theoretical explanations of the phenomena described here.
- Independent publication rights with acknowledgement for researchers who prefer to work alone.

Priority will be adjudicated by timestamped submission. We are particularly interested in the following bounded projects:

1. A classification of the operator's recurrence class (Section 2).
2. A predictive model or inverse-design framework for the encoding-to-LCR mapping (Section 9).
3. An empirical evaluation of EOA-43 as a zero-compression representation in on-device NLP pipelines (Section 8).
4. A cryptographic or information-theoretic analysis of the basis structure (Sections 4 and 8).

For collaboration proposals, contact: bdarghamneurolabs@gmail.com

13. PROGRAM ROADMAP: OUR RESEARCH JOURNEY

The EOA Program documents our research journey into these questions. Parts I through V report what we have found so far. We do not claim to have all answers, and we invite parallel independent work.

Part	Title	Status	Category
0	The Engineering of Alphabets as an Open Frontier	This paper	Programmatic overview and question statement
I	A Rank-Collapsed Deterministic Symbolic Representation with a Convergent Recursive Basis	In preparation [1]	Empirical and descriptive
II	Empirical Cross-Encoding Variation of the Limiting Ratio	In preparation [2]	Empirical and descriptive
III	Transient Structure and Detrended Residuals	Future	Analytical
IV	Boundary Mapping on Downstream Tasks	Future	Applied
V	The Operator beta and Its Recurrence Class	Future	Theoretical

Parts I and II are empirical and descriptive. Parts III and IV are analytical and applied. Part V is theoretical. But none of them, individually or collectively, will answer every question posed in this paper. That is by design. The EOA Program is not a closed system; it is an open frontier.

14. CONCLUSION: THE VALUE OF A WELL-POSED PROBLEM

This paper has posed 43 questions across 10 domains, each summarized in one line in Table 1. We present them as open questions, not as claims of discovery. Some may turn out to be answerable quickly; some may dissolve on closer inspection; and some may need mathematical frameworks that do not exist yet. They all arise from the observations described in Sections 2 through 11.

The EOA Program is not claiming to have answers. We are claiming that the questions matter. Consider what we are looking at: a deterministic operator that turns letter names into convergent sequences, groups those letters into a basis no one can explain, and collapses word vectors down to a line. We have not found any framework we surveyed that accounts for this behavior. Whether that gap is real, or only a gap in our reading, is itself an open question.

We are not experts in every field this touches, so we are asking for help. Mathematicians: what kind of operator is this? Phoneticians: why does the ratio shift when the encoding changes? Representation theorists: is the dimensionality collapse telling us something, or is it just an artifact? Cryptographers: does the mapping have any structural properties worth exploiting? Computer scientists: can you find a task where this actually helps? And to everyone else, tell us honestly whether we have found something deep or merely stumbled into a very persistent coincidence.

The questions are open.

ACKNOWLEDGEMENTS

NO EXTERNAL FUNDING SUPPORTED THIS WORK. WE WROTE THE CODE IN PYTHON, USING NUMPY AND SCIPY FOR THE NUMERICAL COMPUTATIONS.

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